

Validation Rules in a Database

Validation rules in a database are constraints or conditions applied to fields to ensure data accuracy, consistency, and integrity. These rules prevent users from entering incorrect or invalid data.

In MS Access, the LIKE operator is used in Validation Rules to enforce specific patterns for data entry in a field. It allows you to define acceptable formats using wildcards. Wildcards are special characters used with the LIKE operator in queries and validation rules to match patterns in text fields. They allow flexible searches and data validation.

Wildcard	Description	Example in LIKE	Matches
* (Asterisk)	Represents zero or more characters	"A*"	"Apple", "A", "Access"
? (Question Mark)	Represents exactly one character	"A? C"	"ABC", "ACC" (but not "AC" or "ABBC")
# (Hash)	Represents a single digit (0-9)	"###-###-####"	"123-456-7890"
[] (Brackets)	Defines a set of characters to match	"[AB]c"	"Ac", "Bc" (but not "Cc")
[^] (Caret inside Brackets)	Excludes specific characters	"[^AB]c"	"Cc", "Dc" (but not "Ac" or "Bc")
-(Hyphen inside Brackets)	Defines a range of characters	"[A-Z]"	Any uppercase letter (A to Z)

1. Email Address Validation

Like "*@*.*"

"Enter a valid email address (e.g., user@example.com)."

2. Contact Number (10 or 11 Digits) – Only Numbers Allowed

Like "#####" Or Like "#####"

3. CNIC (Pakistani Format: 13 Digits, With Dashes Allowed)

Like "#####-#####-#"

4. Restricting Input to Alphabets (e.g., Names Only)

Not Like "[0-9]*"

5. Date of Birth (Must be 18 or Older)

$\leq \text{DateAdd}(\text{"yyyy"}, -18, \text{Date}())$

6. Record-Level Validation

A person's retirement date cannot be before their date of birth
 $[\text{RetirementDate}] \geq [\text{DateOfBirth}]$

7. Total Marks cannot be more than Maximum Marks

$[\text{TotalMarks}] \leq [\text{MaxMarks}]$